TUM School of Management
Technical University of Munich
Arcisstraße 21
80333 Munich, Germany
aris.dressino@tum.de

September 18, 2026

Professor Roman Słowiński
Co-ordinating Editor-in-Chief
European Journal of Operational Research
Institute of Computing Science, Poznań University of Technology
60-965 Poznań, Poland

Dear Professor Słowiński,

Please find enclosed our manuscript, **"OR-Select: Deterministic Best-Answer Selection over Stochastic LLM Ensembles via Integer Programming,"** submitted for consideration as a Research Article in the *European Journal of Operational Research*.

Large language models sample stochastically, so ensembling N candidate answers to the same query is now common practice, but the standard aggregation rule – self-consistency (majority vote) – is undefined once candidates fail to repeat verbatim, which we show occurs on essentially every open-ended query. OR-Select reformulates best-of-N selection as a multi-objective integer program over an LLM-judged pairwise comparison matrix, aggregated by a Copeland-style statistic that we prove is a 5-approximation to the exact, NP-hard Kemeny-optimal ranking (via Coppersmith, Fleischer, and Rudra's tournament result), with the exact Kemeny-Young solution itself implemented and reported as a reference baseline. The selection step is solved exactly and deterministically with CP-SAT; what is not automatically reproducible – whether the judge's own outputs are stable – is measured directly rather than assumed, and we report a concrete provider-side seed non-determinism failure together with the fix that closes it.

We believe this manuscript is a good fit for *EJOR*'s readership at the intersection of classical rank-aggregation theory and a genuinely new application domain: selection among stochastic outputs of large language models. The paper validates the judge component in isolation against RewardBench (161 human-labeled preference pairs, 90.1% accuracy) before reporting five evaluation tracks and 326 ground-truth rows across four public benchmarks (GSM8K, MATH Level 5, AIME 2026, and GPQA Diamond), comparing OR-Select against self-consistency, greedy single-sampling, and the exact Kemeny-Young baseline throughout. We report the work's limitations directly, including its current smoke-test scale and a self-consistency term that has not yet been load-bearing on any real winner, rather than rounding the evaluation up to a cleaner story.

All authors have approved the manuscript and agree to its submission to *European Journal of*

Operational Research. We have no conflicts of interest to disclose. Thank you for your consideration.

Sincerely,

Aris Dressino